Clinical Nutrition ESPEN 64 (2024) 483–495

Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: http://www.clinicalnutritionespen.com

Narrative Review

Food additives and their implication in inflammatory bowel disease and metabolic syndrome

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A R T I C L E I N F O

Article history:
Received 21 August 2024
Accepted 29 October 2024

Keywords:
Western diet
Emulsifiers
Ultra-processed foods (UPF)
Preservatives
Synthetic colors

S U M M A R Y

Over the past half a century the Western diet (WD) has become saturated with food additives. During the same time, there has been an increase in Western diseases, such as inflammatory bowel disease (IBD) and metabolic syndrome (MetS). Emerging research has shown that food additives may be implicated in these diseases. However, critics have suggested that some of this research is problematic and may cause unnecessary fear amongst consumers. Here we review the emerging research concerning food additives and their implication in IBD and MetS, and criticisms thereof. To make the review more relevant to the WD, we only included common food additives, selected using supermarket data. Over a dozen common food additives from four categories were identified for their potential role in directly promoting these diseases. A consistent limitation of the research was the use of unrealistic human exposure conditions, such as high doses and modes of administration, as well as a lack of human trials. Another limitation was the absence of studies investigating the potential synergetic effect of consuming multiple food additives, as is common in the WD. Despite the limitations, there is some evidence that common food additives may be contributing to these additives, especially via their dysbiotic effect on the gut microbiota.

1. Introduction

In the 1970's, the term Western diseases was first used to refer to several major diseases predominantly observed in Western society, with a notable correlation to the adoption of Western lifestyle and dietary practices [1]. The Western diet (WD) is a post-industrial dietary pattern that is characterised by a high consumption of refined vegetable oils, animal saturated fatty acids (SFA), and refined cereal grains and sugars. The transition to the WD has had an adverse effect on several dietary indicators, such as glycaemic load, caloric density and intake, an imbalance in monounsaturated fatty acid (MUFA), n-3 polyunsaturated fatty acid (n-3 PUFA) and n-6 polyunsaturated fatty acid (n-6 PUFA), sodium and potassium, and deficiencies in micronutrients and fibre [2].

Another dietary practice associated with the WD is a high consumption of ultra-processed foods (UPF), the use of which has increased significantly due to the advances in industrial processes, the centralisation of the global food market and socio-economic changes [3]. As defined by the Food and Agriculture Organization of the United Nations, UPF are formulations of ingredients, mostly of exclusive industrial use, typically created by series of industrial techniques and processes. They often contain multiple ingredients made from high-yield plant foods which have undergone extensive chemical modifications, in addition to ingredients which are rarely

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https://doi.org/10.1016/j.clnesp.2024.10.171

used in domestic cooking, such as food additives. These characteristics make UPF easily consumable, inexpensive and shelf stable [4].

Food additives are ingredients intentionally added to food during various stages, such as preparation, manufacturing, processing, packaging, storage, or transportation, with purpose of modifying the physical, chemical, organoleptic, or biological properties [5]. The hundreds of different food additives currently in use can be classified under their functional categories, such as preservatives, sweeteners, colours, flavourings, fat replacers, nutrients, emulsifiers, stabilisers and thickeners, acidity regulators, leavening agents, anti-caking agents, humectants, and firming agents. Each of the food additives that come under these classifications can be further identified by a specific E-number. For example, E-numbers 100–199 are food colours, whilst culinary ingredients such as spices, herbs, and salt are excluded from this classification system. Most of these food additives are undoubtfully useful for increasing shelf life, food safety, organoleptic properties and are safe to use, as they undergo safety evaluations prior to approval. These safety evaluations consider the immediate and long-term health effects of food additives in the amount typically consumed [5].

Of interest, the emergence of UPF and food additives has coincided with the growing incidence of Western diseases, such as inflammatory bowel disease (IBD) and metabolic syndrome (MetS), which are now major global healthcare concerns [6–8]. Inflammatory bowel disease is characterised by chronic inflammation of the gastrointestinal tract (GIT), expressed in two phenotypes, ulcerative colitis (UC) and Crohn's disease (CD). The pathogenesis of IBD is linked to continues invasion of the intestinal epithelium by harmful microbes and toxins. This may occur due to an increase in pathogenic microbes or a compromised intestinal barrier, thereby eliciting a prolonged immune response [9,10]. Metabolic syndrome is a cluster of metabolic abnormalities such as insulin resistance, manifested as impaired fasting glucose, glucose intolerance, or type 2 diabetes mellitus (T2DM), alongside obesity, hypertension, microalbuminuria, and dyslipidaemia (high triglyceride levels and low high-density lipoprotein (HDL) cholesterol) [11]. Both IBD and MetS share pathogenic pathways involving intestinal barrier dysfunction and chronic inflammation, with strong causal links to gut microbiota disruption [12].

The North American obesity epidemic started immediately after the emergence of UPF. The prevalence of obesity remained stable at approximately 5 % before 1976, but then tripled in the next twenty-five years [13]. During the same period, the prevalence of IBD also increased. The incidence of CD remained stable below 1 per 100,000 prior to the 1960's, but then accelerated to 7 per 100,000 over the next twenty-five years [7]. A similar trend occurred with the prevalence of diagnosed diabetes, which was stable prior to 1960 at approximately 1 %, but then increased to 2.6 % over the next twenty-five years [14]. Furthermore, multiple reviews have shown a positive correlation between UPF consumption and non-communicable diseases. These associations are likely due to the already mentioned characteristics of the UPF saturated WD [4,15–17]. However, an overlooked component of the WD that may be contributing to these diseases are food additives.

The consumption of food additives is increasing globally, with some evidence suggesting that food additive consumption is outpacing the rise of UPF consumption [18,19]. Whilst food safety government bodies like the U.S. Food and Drug Administration (FDA), the European Food Safety Authority (EFSA) and the Food Standards Australia New Zealand (FSANZ) regulate the use of these additives, a growing number of evidence is challenging their safety [20–24]. However, critics have also suggested that some of recent evidence is problematic and may cause unnecessary fear amongst consumers [25–27]. Therefore, we will review the research concerning food additives and their direct association with IBD and MetS, and the criticisms thereof. Since there are hundreds of additives currently in use [5], we will first conduct a screen using supermarket data to select common food additive categories to review.

2. Food additives

A recent comprehensive study used food sales data from eighty countries to evaluate the global trend in UPF consumption, including food additives. Total food additives supplied from UPF were 2.3 kg, 0.7 kg and 0.2kg/capita in high, middle, and low-income countries, respectively [3]. To identify which are the most prevalent food additive categories used in the WD, data from supermarkets in four countries (United States of America (US), Australia (Aus), France and Brazil) were summarised (Fig. 1). These countries represent a range of UPF consumption levels (highest to low intake of UPF per capita, respectively) at the same compounding annual growth rate, thus offering a glimpse into the global food additive usage [3]. Moreover, these four countries have previously conducted surveys to investigate food additives in supermarket groceries [18,19,28,29].

Surveys from these countries revealed varying patterns of food additive usage. A Brazilian survey found the most common foods were sweets, processed meats, and convenience foods, such as frozen pizza, French fried and noodles [28]. Over 94 % of sweets had at least one food additive, and 36 % had over six food additives per item. Approximately 85 % of processed meats had at least one food additive, and 37 % had over six food additives per item. Finally, 77 % of convenience foods had at least one food additive, and 25 % had over six food additives per item. Flavourings were the most common food additive and were found in 47 % of all food items, followed by preservatives, colours, stabilisers and emulsifiers, at 29 %, 28 %, 28 % and 19 %, respectively.

Similarly, a French survey found the most common foods were sweets and pastries, one-dish meals and processed meats [18]. Emulsifiers were the most common food additive and were found in approximately 28 % of all food items, followed by flavourings and preservatives, at 21 % and 16 %, respectively. Food items most likely to have food additives were also the ones most available. Overall, 54 % of all food items had at least one food additive and 36 % had at least two. An Australian survey showed that flavourings were the most common food additive, then colours and emulsifiers, at 31 %, 18 % and 16 %, respectively [29].

The US study used more direct methods for their survey and measured food additives in food items purchased over time, thus producing more robust results [19]. The survey found that sweets and snacks were the most purchased food items and contained an average of six food additives per food item. These food items contained five food additives per item two decades prior, showing an increase in the use of food additives. Furthermore, there was a decrease in the purchases of food items containing no food additives, from 45 % to 24 % over two decades. The most common food additives were colours, flavourings, and preservatives, at 47 %, 39 % and 38 %, respectively. However, since emulsifiers were not measured another US dietary survey on emulsifier intake may be supplemented [26]. This survey found that emulsifier consumption was ubiquitous in the US population, with 84 % of participants consuming the seven most common emulsifiers over a two-day period.

Using the supermarket produce availability surveys, it is reasonable to assume that flavourings, colours, preservatives, and emulsifiers, are the most common food additives in supermarket food items and in the WD. As such, only additives from these four categories will be reviewed. Whilst this is not a direct measure of food additive consumption, it is a common proxy for population

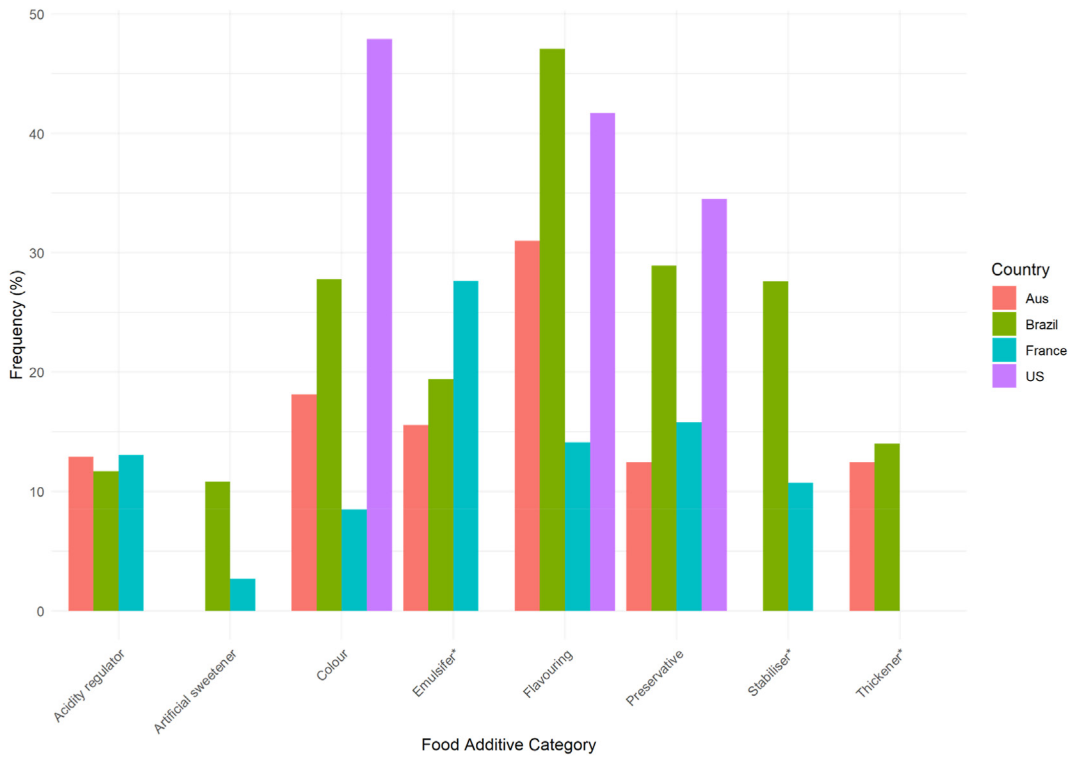

Fig. 1. The frequency of food additive categories used in supermarket food items across four countries: Australia (Aus), United States of America (US), France and Brazil. *Of note, emulsifiers can be categorised as stabilisers and thickeners. Data sourced from [18,19,28,29]. Made with R Core Team (2024): A language and environment for Statistical Computing.

exposure to food additives [18,19,29,30]. This is partly because supermarkets are the primary suppliers of food items and proportionally reflect consumer demand in high-income countries [3,31].

3. Emulsifiers

Emulsifiers are food additives that are commonly used in the food industry to form and maintain emulsions between immiscible food ingredients, such as water and oil. Emulsifiers can be classified into two main categories, surfactants, and amphiphilic biopolymers. A surfactant type emulsifier is comprised of a hydrophobic tail and a hydrophilic head, which can be cationic, anionic, non-ionic or zwitterionic, facilitating the formation of stable oil-water interface. Examples of such emulsifiers are ethoxylated sorbitan esters (P20, P60, P65 and P80) and lecithin. Amphiphilic biopolymers have a similar effect but a different structure, and may be proteins, polysaccharides, or a combination of both. Examples of these are xanthan gum, modified starch, and pectin. Emulsifiers may also have multiple food additive functions. Modified celluloses such as carboxymethyl cellulose (CMC) and carrageenan (CGN) are used as emulsifiers, but also as thickening and gelling agents [25]. To simplify the nomenclature, emulsifiers with multiple functions will be classified as emulsifiers from here on in.

Furthermore, on classification, an additive with a single E number may come in different forms. The emulsifier carrageenan is a sulphated polygalactan derived from red seaweed. It is a polymer of repeating galactose groups with attached sulphate groups, the position of which determine whether CGN is categorised as kappa (κ), iota (ι) or lambda (λ). Carrageenan can also be differentiated on its molecular weight (MW). Lower MW CGN, also known as degraded CGN (dCGN) or poligeenan, may have a MW between 10 and 20 kDa. Higher MW CGN, which is preferred by the food industry, has a MW above 100 kDa, but more often between 200 and 800 kDa [27].

Although the presence of emulsifiers, and other food additives, in the food industry is well established it is difficult to quantify their actual consumption. This is partly due to a lack of data on food additive concentrations in products, ingredients labelling legislation, and an overall shortage of dietary surveys that include food additive estimates [32–34]. One of the largest and most accurate dietary surveys to date to include food additives was conducted on the French population [35]. Based on the responses from 106,000 participants, it was estimated that the most frequently consumed emulsifiers were modified starches, xanthan gum, pectin, mono-diglycerides (MDGs), lecithin and CGN, with average intakes of 24.3 mg, 6,6 mg, 3.0 mg, 2,4 mg, 0.8 mg, and 0.7 mg per kg_{bw}/day, respectively. Over 77 % of participants consumed all these emulsifiers at least once over two 24-h testing periods. Another study, conducted in the US, showed that the most consumed emulsifiers were lecithin, MDGs, CMC, P80, sodium stearoyl lactylate (SSL), calcium stearoyl lactylate (CSL), sucrose esters, and polyglycerol polyricinoleate (PGPR) at 61 mg, 75 mg, 24 mg, 5 mg, 7 mg, 7 mg, 9 mg, and 1 mg per kg_{bw}/day, respectively [26]. Of all these emulsifiers, CGN, P80 and CMC have received considerable attention over safety concerns.

3.1. Emulsifiers and evidence of their adverse effects

The harmful effects of dCGN have long been established with earlier studies using it to induce UC in murine models [36]. However, most of the earlier studies used dCGN as a chemical agent for inducing specific pathologies in murine models for research purposes, such as studying IBD [36–38]. Later studies, investigating the pathways of CGN induced IBD, concluded that CGN may be causing IBD like symptoms in humans [39,40]. It was also shown that CGN induces inflammation in human epithelial cells using the same toll-like receptor 4 (TLR4) biochemical pathway as lipopolysaccharides (LPS) in a dose-dependent manner [40]. Furthermore, CGN may directly disrupt human epithelial cell tight junctions, which can exasperate IBD like symptoms by weakening the intestinal epithelium barrier [41,42].

Despite criticisms that some studies investigating CGN have not used food grade high MW CGN [25,27], there was sufficient evidence to suggest that CGN may be complicit in IBD and possibly MetS to warrant a double-blind clinical study. Results showed that CGN intake may be associated with hyperglycaemia, insulin resistance and inflammation in prediabetics [23]. Whilst this study showed the harmful effects of CGN using human trials, most of the evidence that CGN may cause IBD is based on murine models. It has been suggested that the results from murine studies may not be relevant to humans because CGN was administered through drinking water, therefore CGN was not bound to proteins, as it would in regular human intake, and could alter the results [25,27]. To address this critique, it would be worth investigating whether the intake of CGN, and possibly other emulsifiers, have a different health effect when they are administered through drinking water or as part of an emulsion in food.

Other emulsifiers have also been the focus of scrutiny. A study by Swidsinski et al. (2009) was one of the first to suggest that the consumption of emulsifiers may be a factor in the rising incidence of IBD, and that this may be caused by their cleaning effect. Results showed that the emulsifier CMC caused symptoms akin to CD in IL-10 deficient mice. Another study then investigated the emulsifier P80 and found that it increased the translocation of a CD's mucosa-associated *Escherichia coli* through human epithelial cells [44]. Chassaing et al. (2015) showed that both CMC and P80 promote MetS and UC, although not through the proposed cleaning effect. Both CMC and P80 caused low grade inflammation, impaired glycaemic control, altered the microbiota, and increased bacterial encroachment and mucosal permeability. However, the emulsifiers had no effect on germ free mice, suggesting that the emulsifiers caused MetS and UC through gut microbiota manipulation.

After that landmark study came an influx of new research on the safety of emulsifiers (Table 1). This included a follow up study by Chassaing et al. (2017) that showed these emulsifiers can directly manipulate the human gut microbiota independent of a host by changing the relative abundance of bacteria and gene expression. However, emulsifier-microbiome research is still in its infancy, new evidence is suggesting that these emulsifiers may also be directly impacting the mucus layer as originally suggested [46,47]. Furthermore, several studies have shown that besides P80 and CMC, most common emulsifiers have a negative effect on the gut microbiota, except lecithin and MDGs, and require further investigation [48,49].

It is important to recognise that recent criticisms have emerged of these emulsifier studies, suggesting that they may be irrelevant to the WD as the doses used far exceed those consumed by humans [25,26,70]. For example, as shown in the previous section, an average person in the US is estimated to consume 5 mg of P80 per kg_{bw}/day. Meanwhile, a mouse in a typical emulsifier study consumes approximately 5 ml of a 1 % emulsifier solution, at an average bodyweight of 23 g [71]. This equates to 2174 mg per kg_{bw}/day, the equivalent to feeding a typical US consumer 185 g/day. This would be an improbable daily exposure. However, it is important to mention that the reason the 1 % dose was initially chosen was because that is the concentration at which P80 and CMC are allowed to be used in food items [20].

The choice of doses in emulsifier studies is further complicated because the same emulsifier can cause completely different results depending on what dose is used. Jiang et al. [2018] exposed mice to 150 mg per kg_{bw}/day of GML, which caused an increase in bodyweight, hyperlipidaemia, inflammation and a 60-fold increase in serum LPS. A follow up study showed that mice exposed to 1600 mg per kg_{bw}/day of GML lowered inflammation and serum LPS, lowered LDL and increased HDL, and improved glucose homeostasis [56]. Therefore, to determine if emulsifiers are complicit in Mets and IBD, it would be informative to test different doses that reflect average human intakes.

4. Colours

Food colours are dyes, pigments or any substances that can impart colour to food items. They are used to enhance or sustain the visual properties of food items, which may have been compromised during the manufacturing process, and are an important component of food item consumer recognition. Food colours can be classified into three basic groups, natural, synthetic, and inorganic. Natural food colours are obtained from animals or plants. A common group of natural food colours are anthocyanins, which are derived from berries and grapes, and take on colours ranging from purple to blue. Natural food colours are generally considered safe and may even have beneficial health effects [72,73]. Synthetic colours, such as azo dyes, are produced by chemical synthesis and have several advantages over natural colours, such as higher colour intensity, stability, uniformity, resistance to light and pH alteration, and cost effectiveness. Inorganic colours are a less common and diverse range of food colours, and are derivatives of titanium (Ti), iron (Fe), silver (Ag), aluminium (Al), and gold (Au) [73,74].

An Australian survey conducted by FSANZ [34] estimated that the most frequently consumed food colours of concern were synthetics: tartrazine, sunset yellow, azorubine, brown HT, brilliant blue and Allura red, with an average consumption of 1.1 mg, 1.1 mg, 1.0 mg, 0.9 mg and 0.5 mg per kg_{bw}/day. These intakes were higher for children aged 6–12 years old, with intakes of 1.3 mg, 1.3 mg, 1.2 mg, 0.9 mg, 0.6 mg, and 0.5 mg per kg_{bw}/day, respectively. A US study also found that tartrazine, Allura red and sunset yellow were the most common food colours used in food items and consumed by the population [75]. The most common inorganic food colour is titanium dioxide (TiO_2), with an estimated US intake of 0.2–0.7 mg per kg_{bw}/day [76]. Food grade TiO_2 comes in the anatase crystallised form with an optimal size between 200 and 350 nm, in which TiO_2 creates its designated white colour. However, particle size may vary depending on the quality of the product, with median particle size reaching 102 nm in lower quality TiO_2, such as E171-E [77,78].

4.1. Colours and evidence of their adverse effects

The safety of synthetic food colours has been under scrutiny since the 1970's because of their association with asthma, allergies, and behavioural disorders in children [79]. Regarding IBD or MetS, most of the research on synthetic food colours has largely focused on the microbiota's role in metabolising food dyes [80] with little evidence currently suggesting that synthetic food colours are complicit in either disease [73,74]. However, several recent studies have shown that tartrazine may cause endocrine disruption and hyperglycaemia, even at doses similar to the acceptable daily intake (ADI) [81,82]. A study by Elwan et al. [2019] also showed that tartrazine may cause lesions in the gastric mucosa via oxidative stress, but at doses far exceeding normal human intake levels. Interestingly, one study showed that the in vivo effect of tartrazine is not replicated *in vitro* [82]. A possible explanation for this difference is that the deleterious effect of tartrazine is caused by its metabolites, such as sulfanilic acid, which are derived from the gut microbiota's metabolism of tartrazine [82–84]. Therefore, future studies should

Table 1
The effects of emulsifiers on metabolic syndrome (MetS) and inflammatory bowel disease (IBD) associated symptoms.

Emulsifier	Model	Dose	Effect	Source
λ-CGN	*In vitro*	0.0001 %	Inflammation	[50]
λ-CGN	*In vitro*	0.0001–0.001 %	Intestinal cell necrosis & reduced cell proliferation	[39]
λ-CGN	*In vitro*	0.0001–0.001 %	Inflammation	[40]
P80	*In vitro*	0.01–0.1 %	*E. coli* adhesion and translocation	[44]
P80 & CMC	*In vitro*	0.1–1.0 %	Microbial compositional changes & increased flagellin	[45]
κ- λ- ι-CGN	*In vitro*	0.20 %	Decreased gut proteolysis	[42]
		0.005–0.00005 %	Gut permeability & dose-dependent but λ- ι- only	
P80 & CMC	*In vitro*	0.5–1.0 %	Structural changes to mucus layer & *E. coli* translocation	[46]
CMC, P80, lecithin, SLs & RLs	*In vitro*	0.005–0.05 %	Dysbiosis, decreased bacterial cell count, decrease in SCFA, increase in pathogens and their virulence	[48]
AMG, GMO, GMS, PGMS & SSL	*In vitro*	0.0008–0.025 %	Increase in Enterobacteriaceae & *E. coli*, & direct inhibition of beneficial bacteria	[51]
P80, CMC & β-lac	Ex vivo	0.002–1.000 %	Increase in gut permeability and cytotoxicity at higher concentrations (P80 at 0.75 %>)	[47]
20 emulsifiers	*In vitro*	1 %	18/20 emulsifiers caused microbial compositional changes, increase in LPS, flagellin and virulence	[49]
EMI & EMII	*In vitro*	0.0005–0.0015 %	Intestinal cell proliferation & inflammation	[52]
dCGN (L)	Murine	5 %	UC symptoms	[36]
λ-dCGN (L)	Murine	1.50 %	UC symptoms	[37]
λ-CGN (L)	Murine	2 %	UC symptoms	[38]
CMC (L)	Murine	2 %	CD symptoms, bacterial overgrowth & encroachment	[43]
ι-dCGN (L)	Murine	5 %	UC symptoms & macrophage infiltration	[53]
P80 & CMC (S/L)	Murine	0.1–1.0 %	Hyperglycaemia, insulin resistance, inflammation, dysbiosis & gut permeability	[20]
κ- λ-CGN (L)	Murine	0.01 %	Hyperglycaemia, insulin resistance & inflammation	[41]
P80 (L)	Murine	1 %	Hyperglycaemia, insulin resistance, inflammation, gut permeability & encroachment	[54]
GML (S)	Murine	0.15 %	Inflammation, less SCFA, hyperlipidaemia, weight gain, increased HDL, dysbiosis	[55]
GML (S)	Murine	1.6 %	Reversed HFD induced dysbiosis, MetS and inflammation	[56]
P80 (L)	Murine	1 %	Inflammation, reduced microbial diversity & increase in sulphide producing bacteria	[57]
P80	*In vitro*	0.02 %	Increased pathogenic bacteria swarming	
P80 (L)	Murine	1 %	Weight gain, mild UC, decrease in GLUT1 and synergistically exasperated epithelial damage post radiation therapy	[58]
P80 & CMC (L)	Murine	1 %	Inflammation, microbial compositional changes, weight gain, hyperglycaemia & increased *E. coli* virulence	[59]
P80 (L)	Murine	1 %	Impaired insulin sensitivity & glycolytic metabolism, & increased intestinal permeability	[60]
P80 (L)	Murine	1 %	Maternal exposure increased offspring susceptibility to UC	[61]
P80 & CMC (L)	Murine	1 %	Inflammation and histological UC, microbial compositional changes & increased virulence	[62]
P80 & CMC (L)	Murine	1 %	Small intestine tumours, inflammation and increase in LPS (sex dependent)	[63]
κ-CGN (L)	Murine	41.7 mg/kg	Decreased SCFA bacteria, increased mucus degrading bacteria, exasperated inflammatory response to pathogen	[64]
P80 (L)	Murine	1 %	Obesity, impaired intestinal development, loss of tight junctions, decreased MUC2 expression, inflammation & change in bile acid composition	[65]
GMD (L)	Murine	0.15–1.6 %	Weight gain & hyperlipidaemia, decreased inflammation & serum LPS, increase in tight junctions, SCFA & MUC2 expression	[66]
SSL & P20	Poultry	0.05 %	Weight gain	[67]
CGN	RCT	200 mg/day	Promotion of UC relapse	[68]
CGN (S)	RCT	250 mg/day	Improved insulin homeostasis and inflammation in CGN free group	[23]
CMC (S)	RCT	15 g/day	Bacterial encroachment & LPS increase in CMC sensitive participants only	[69]

UC, ulcerative colitis, CD, Crohn's disease, SCFA, short-chain fatty acid, dCGN, degraded carrageenan, CGN, carrageenan, (L), dissolved in liquid, (S), dissolved in solid, κ, kappa-carrageenan, ι, iota-carrageenan, λ, lamda-carrageenan, CMC, carboxymethyl cellulose, P80, ethoxylated sorbitan ester 80, GML, glycerol monolaurate, SSL, sodium stearoyl lactylate, P20, ethoxylated sorbitan ester 20, SLs, sophorolipids, RLs, rhamnolipids, AMG, glycerol monoacetate, GMO, glycerol monooleate, GMS, glycerol monostearate, PGMS, propylene glycol monostearate, SSL, sodium stearoyl lactylate, β-lac, β-lactoglobulin, GMD, glycerol monodecanoate.

investigate the relationship between the microbiome and the indirect harmful effect of tartrazine.

A recent study has implicated another azo dye in relation to IBD. Kwon et al. [2022] investigating the effect of synthetic food colours on intestinal serotonin showed that Allura red enhances the susceptibility to dextran sulfate sodium (DSS)-induced UC. Mice fed human equivalent doses of Allura red had elevated levels of serum and faecal lipocalin-2 and had reduced MUC2 expression. These physiological changes were not present in serotonin deficient mice, suggesting that Allura Red can induce UC via the serotine pathway. Furthermore, Allura Red exposed gut microbiota transplanted into germ free mice increased the severity of DSS-induced UC, suggesting a microbiota-dependant pathway.

Most evidence of a food colour causing IBD comes from research done on the inorganic colour TiO_2. After ingestion, TiO_2 passes through the GIT intact, although the physiochemical properties of the GIT cause particle agglomeration [85]. Whilst in the GIT, TiO_2 can interact with the microbiota, causing structural changes such as reducing the number of probiotic genera, *Lactobacillus*, *Bifidobacterium*, and *Barnesiella*, and shifting the *Firmicutes-Bacteroidetes* ratio. These changes are associated with a reduction in MUC2 and SCFA production, and an increase in pro-inflammatory cytokines and neutrophil infiltration [86–90]. Higher doses, exceeding normal human intake, have also shown to cause histological abnormalities in intestinal epithelial cells and elevate serum triglycerides and glucose [85,89,90]. The adverse health effects in these studies are generally more severe as particle size decreases.

However, the effect of TiO_2 on the gut is not always reproducible, with several authors suggesting that for TiO_2 to have an adverse effect on gut health there needs to be prior disturbance of the intestinal barrier. The disturbance increases intestinal permeability and promotes the absorption and accumulation of TiO_2, which may then lead to IBD like symptoms [87,89,91,92]. This may explain why elevated serum TiO_2 levels have been detected in active UC patients but not in those in remission [93].

Due to accumulating evidence, the EFSA recently conducted a safety re-evaluation of TiO_2 and concluded that it may no longer be considered safe as a food additive [92]. There was sufficient evidence to show that TiO_2 may promote inflammation and aberrant crypt foci in the colon, which was previously demonstrated by Bettini et al. (2017). However, FSANZ responded to the EFSA restrictions, stating that there is not enough evidence that food grade TiO_2 is of concern to human health and that studies on inflammation are rarely reproducible, even when testing doses as high as 1000 mg per kg_{bw}/day. A major critique of the evidence was that in most murine studies test animals were administered TiO_2 in aqueous solutions, either by oral gavage or drinking water, which required sonication dispersion of TiO_2. This method is not used by the food industry because TiO_2 is generally used as part of a food matrix. The alleged consequence of this laboratory protocol is a reduction of particle size, production of free radicals in the solution, and decreased TiO_2 agglomeration in the gut [78].

5. Flavourings

There are currently over 1700 different flavouring compounds available in the food market which are made from complex mixtures of natural and synthetic components [5]. Natural flavourings are derived from animal and plant sources and may include oils, spices, and the end products of distillation, fermentation, roasting or heating. Synthetic flavourings are produced by chemical synthesis and have not been identified in natural products. However, they may have chemical structures identical to natural flavourings. For example, the oldest industrial flavouring vanillin, which is the primary component responsible for the vanilla flavour, can either be isolated from vanilla extract or be chemically synthesized from the organic compound guaiacol [94,95]. Flavourings also include flavour enhancers, which do not contribute their own flavour but rather enhance the flavour of the food [5].

The most common food additive and flavouring is citric acid, as a flavouring, it gives a sour, tart, and acidic flavour, and can enhance the perceived sweetness of foods [35,96]. A French survey found that 91 % of consumers ingest citric acid almost daily, at an average of 30 mg per kg_{bw}/day [35]. Some earlier estimates have found citric acid intake to be as high as 400 mg per kg_{bw}/day [97]. The second most common flavouring is the flavour enhancer monosodium glutamate (MSG). Monosodium glutamate is produced through fermentation using the bacterium *Corynebacterium glutamicum* and is the salt of glutamic acid responsible for producing the umami flavour. It is the most consumed food additive by weight, with French consumers ingesting an average of 36.5 mg per kg_{bw}/day [5,35]. This estimate is likely to be similar in other European and Asian countries [98,99].

The intake of other flavourings is difficult to determine. In part, this is because flavourings that are comprised of multiple ingredients are only labelled as their intended flavour so the same labelled flavour may be manufactured using different ingredients [95]. For example, the European intake of smoke flavouring is 23.8 mg per kg_{bw}/day. This estimate is derived using sixteen different products, each comprised of different ingredients [100].

5.1. Flavourings and evidence of their adverse effects

As there is no current evidence to suggest that citric acid consumption has any adverse health effects, even when used at doses a thousand-fold the estimated daily intake [101], the only commonly consumed flavouring that may be relevant to IBD or MetS is MSG. After ingestion, MSG is mostly metabolised by enterocytes for energy, with partial absorption into the portal system or conversion into other amino acids [102,103]. Claims of adverse health effects associated with MSG began in 1969, initially labelled as the Chinese restaurant syndrome. Whilst the initial adverse health claims were dismissed, either due to problematic experimental designs or over exaggerations of rare allergic reactions [104], new research has shown that MSG may be linked to MetS.

Monosodium glutamate has been shown to cause insulin resistance, dyslipidaemia, fatty-liver disease, hyperglycaemia and T2DM in murine models, with one study concluding that there is sufficient evidence for MSG to be removed from the diet [105–108]. These studies have been criticised for administering MSG through subcutaneous injections, which has little relevance to dietary human exposure as it bypasses the normal metabolic pathways [109]. However, these studies did influence the study by Collison et al. (2009), which used orally administered MSG at 91 mg per kg_{bw}/day. After thirty-two weeks, the test animals had elevated levels of serum insulin and triglycerides, with an increase in abdominal adipose tissue, despite no weight gain. Furthermore, there were significant changes in hepatic gene expression, with MSG activating genes involved in lipid storage, oxidation, and the clearance of exogenous lipids.

Since the animal models showed a link between MSG intake and MetS, a human study was conducted [98]. After a ten-day consumption of approximately 57 mg per kg_{bw}/day of MSG, results showed that there was a correlation between MSG intake and MetS criteria. It was estimated that the risk of developing MetS increased 1.14-fold for each gram of daily MSG consumption. When single criteria of MetS were analysed, there was significant trend towards insulin resistance. A follow up study using rats showed that MSG consumption suppressed beneficial gut bacteria and promoted bacteria associated with MetS, in addition to changes to metabolic pathways [110]. Of interest, these studies have been criticised by the International Glutamate Technical Committee for not accounting for baseline glutamate intake and using unrealistic doses [111,112]. Moreover, the food company that manufactures MSG replicated the study by Collison et al. (2019) and concluded that there insufficient evidence to show that MSG promotes MetS [113]. These criticisms and the controversial history have complicated the research on MSG safety, with further investigation required to confirm if MSG may be complicit in MetS.

6. Preservatives

Preservatives are substance that are added to food items to increase their shelf life by reducing water availability, slowing down chemical reactions, and inhibiting microbial growth. Since microbial spoilage is one of the biggest challenges in food preservation most preservatives act as antimicrobial agents. Some preservatives such as salt and sugar have been used in day-to-day cooking prior to the emergence of UPF and will therefore be excluded from this review. The remaining preservatives can be classified into several groups: The weak organic acids such as benzoates, propionates, and sorbates and salts thereof, the inorganic ones such as sulphites and nitrites, and the P-hydroxybenzoic acid alkyl esters (parabens) [5,114].

The intake of benzoates (E210-213), propionates (E281-282) and sorbates (E200-203) has been estimated to be between 0.5 and 8.4 mg, 7.7–21.7 mg and 4.7–23.7 mg per kg_{bw}/day, respectively [115–117]. Of note, the gut microbiota can produce propionate by fermenting non-digestible carbohydrates, but the preservative propionate form is almost entirely produced by synthetic petrochemical routes [116]. The intake of sulphites (E220-228) has been estimated to be 2.21 mg per kg_{bw}/day. Whilst this intake is lower by weight than the previous three preservatives, sulphites are often consumed above their ADI of 0.7 mg per kg_{bw}/day [118,119]. Moreover, earlier surveys from the US have estimated the average intake to be 19 mg per kg_{bw}/day [120]. The intake of nitrites (E249-240) has been estimated to be between 0.03 and 0.05 mg per kg_{bw}/day. Natural sources and environmental contamination are the major sources of nitrite intake, with food additive nitrite contributing 17 % to total intake. Despite this low intake of nitrite by weight its extensive use in processed meat makes it one of the most frequently consumed preservatives [35,121]. Finally, the intake of parabens (E209,216, 214 and 218) has been estimated to be 1.29 mg per kg_{bw}/day, but only 0.04 mg of these parabens come from dietary sources [122], and therefore are unlikely to be associated with the WD.

6.1. Preservatives and evidence of their adverse effects

Since the primary role of preservatives is to inhibit the growth of microorganisms, the most obvious mechanism by which preservatives may promote IBD or MetS is through their effect on the microbiome. However, it has been pointed out that research on the effect of preservatives on microorganisms has predominantly been conducted in the context of food safety, with little research done on the impact of preservatives on the gut microbiota [123]. The few studies that have pursued this line of enquiry have shown that preservatives significantly affect the gut microbiota. Moreover, beneficial bacteria are often more susceptible than pathogens to the antimicrobial effects of preservatives [24,123–125]. The first study to investigate the impact of preservatives on the human gut microbiota showed that benzoate, nitrite, and sorbate cause dysbiosis, and compositional changes associated with IBD. Of importance, the doses used in this study were within realistic human intake: benzoate, nitrite and sorbate at 4.8 mg, 0.36 mg, and 19 mg per kg_{bw}/day, respectively [125].

The dysbiotic effect of benzoate, nitrite, and sorbate has been replicated, in addition to propionate and parabens. These preservatives were shown to cause a reduction in the beneficial genus *Bifidobacterium* and enrichment of the phyla *Proteobacteria* [24], which has been shown to promote inflammation and is associated with IBD and MetS [126]. Results also indicated that chronic exposure to these preservatives impacts glucose metabolism, with benzoate and propionate inducing hyperglycaemia after eight weeks [24]. Another study showed that ingested propionate causes hyperglycaemia when administered to mice using normal human intake doses [127]. It was demonstrated that after ingestion propionate enters the circulatory system and stimulates the release of glucose producing hormones via excitation of the sympathetic nervous system. Results were verified in a follow-up randomized placebo-controlled study. Importantly, the authors suggested that orally ingested propionate does not mimic the beneficial effects attributed to microbiome derived propionate [127,128].

The last group of preservatives of interest are sulphites, which have previously received attention for inducing allergy like symptoms in sulphite sensitive-people, and for uncertainties over current ADIs [118,119,129]. Regarding IBD, sulphites have been shown to cause gastric lesions in animal models as far back as 1972. The results of these earlier studies were used to establish the current sulphites ADI [118,130]. More recent studies have further shown that sulphites can inhibit beneficial bacteria, cause mucosal cell death via oxidative stress, and induce UC in animal models [124,131,132]. However, the gastric lesion effect of sulphites is not always reproducible [133] and the doses used in these earlier studies exceed the estimated daily intake of sulphites, therefore, further studies are still required to determine whether sulphites are complicit in IBD.

7. Food additives and dysbiosis

This review has highlighted several different pathways by which food additives may be promoting IBD and Mets. One pathway, that is ubiquitous amongst most these additives, is their effect on the gut microbiota. It has now been well established that bacterial dysbiosis is involved in the pathogenesis of multiple diseases [12]. Patients with IBD exhibit a reduced bacterial diversity, elevated abundance of Enterobacteriaceae and *Proteobacteria,* and a reduced abundance of *Bacteroidetes* [134,135]. A reduction in diversity is considered the most reproducible finding from IBD microbiome studies [136]. Furthermore, it has been shown that specific bacterial species have a disproportionate contribution to IBD and MetS, these include the elevated abundance of *Escherichia* spp, *Fusobacterium* spp, *Ruminococcus* spp, and a decrease in SCFA producers such as *Faecalibacterium prausnitzii* and *Bifidobacterium* spp [12,134,137]. As observed in Table 2, the reviewed food additives replicate these gut microbial shifts.

The decreased abundance of SCFA producers and the increased abundance of mucolytic bacteria may compromise the intestinal barrier, thus allowing entry of pathogens and bacterial products, such as LPS, into circulation and thereby inducing chronic low-grade inflammation. This is thought to be an early event in IBD pathogenesis [12]. Moreover, a compromised intestinal barrier, in conjunction with elevated LPS, induces insulin resistance via activation of the TLR 4 proinflammatory pathway [12,138]. Therefore, one of the most common pathways by which food additives may be promoting MetS and IBD is through their alteration of the gut microbiota towards a proinflammatory state.

8. Discussion and conclusion

The emergence of UPF has aided food affordability and shelf-life for many, however, it has to be recognised that this has resulted in an increase in the intake of food additives over the last half a century [18,19]. People in high-income countries are consuming approximately 6.3 g of food additives per day and the use of food additives is increasing, with the most purchased foods now containing an average of six food additives per item [3,19]. Furthermore, high UPF consumers may be ingesting considerably higher amounts since it is common for popular fast food restaurants to serve food items that can contain over twenty different food additives per food item [160]. During this time period, the prevalence of Western diseases such as IBD and MetS has accelerated [7,13,14]. Whilst several recent dietary changes, such as a higher intake of calories and a reduced intake of fibre, have been associated with these diseases [1,2,136,161,162], an overlooked component of UPF that may be contributing to these diseases are food additives.

Using supermarket data, we identified over a dozen common food additives from four categories which may be contributing to IBD and MetS. Of the four food additive categories summarised in this review, emulsifiers may be the most problematic. The influx of

Table 2
The *in vitro* and *in vivo* alterations of the gut microbiota by common food additives, and their implication in disease.

Food additive	Gut microbiota alterations		Microbial function and relevance to disease
	Increase	Decrease	
P80, CMC, RLs, SLs, Benz, Sorb		Bacteria diversity [20,48,57,125]	Reduction in diversity is a hallmark of dysbiosis and IBD [12,135,136]
MSG, Sorb, Benz, CGN		*Firmicutes* [125,139,140]	Dominant anti-inflammatory phylum of the gut [12,141], which is reduced with IBD [12] but elevated with obesity [134]
TiO_2	*Firmicutes* [87,90]		
TiO_2	*Actinobacteria* [142]		Dominant phylum of the gut involved in lipid metabolism, vitamin biosynthesis, gut homeostasis and SCFA production [143]
TiO_2, Benz, Prop, Sorb, Sulp, P80, CMC, CGN		*Actinobacteria* [24,63,87,140]	
CGN,		*Proteobacteria* [64]	Dominant proinflammatory phylum [126] of the gut which is elevated with MetS and IBD [12]
P80, CMC, TiO_2, Benz, Prop, CGN	*Proteobacteria* [20,24,125,140,142,144]		
CGN, P80, TiO_2	*Bacteroides* [54,64,85,142]		Dominant beneficial phylum associated with SCFA production, and glucose and lipid metabolism [145], which is reduced with diabetes and IBD [12]
P80, RLs, CGN, SLs, TiO_2		*Bacteroides* [48,57,87,89,140]	
Benz, Sorb	*Verrucomicrobia* [125]		Decreased dominant phylum in dysbiosis [12]
AR		*Verrucomicrobia* [146]	
MSG	*Clostridium* [110]		TMA producer [147]
P80	*Clostridium* cluster XI [54]		Secondary bile acid producers [148]
CGN, SSL, MSG, TiO_2, Benz,		*Bifidobacterium* [24,51,64,87,110,149]	Probiotic associated with weight loss [12,150], SCFA production, and decreased inflammation and MetS [12,151]
CGN,	*Bacteroides ovatus* [64]		Mucolytic bacteria [64]
CGN, P80, CMC, TiO_2		*Akkermansia mucinphila* [20,64,90]	Probiotic [152,153]
CGN		*Faecalibacterium prausnitzii* [64]	SCFA producer [154], reduces colitis [134]
CGN		*Lachnospiraceae* [64]	SCFA producer [154]
CGN		*Ruminiclostridium* [64]	SCFA producer [154]
P80, CMC		*Bacteroidales* [20]	Reduction associated with IBD [12]
P80, Benz, Prop, Sulp,	*Helicobacter* [24,54]		Pathobiont [137] and TMA producer [147]
P80	*Campylobacter jejuni* [54]		Pathobiont [155]
P80	*Salmonella* spp [54]		Pathobiont [156]
CGN, P80, CMC, SSL, MSG	*Ruminococcus* [20,51,64,139]		Mucolytic bacteria [157] associated with IBD and MetS [12]
TiO		*Barnesiella* [85,90]	Inhibits pathogens [90]
P80, RLs, SLs, SSL, GMS	*Enterobacteriaceae* [48,51,57]		Sulphide producer [57] associated with IBD [136,137] and MetS [12]
RLs, SLs, SSL, TiO_2	*Escherichia/Shigella* [48,51,85]		Proinflammatory pathobiont associated with IBD [44,158], MetS and NAFLD [12]
RLs, SLs,	*Fusobacterium* [48]		Associated with MetS [12] and CRC [137]
RLs, SLs,	*Acidaminococcus* [48]		Associated with NAFLD [12]
CGN, TiO_2		*Prevotella* [64,85]	SCFA producer [159]
MSG	*Roseburia* [139]		SCFA producer [12]
TiO_2		*Lactobacillus plantarum* [149]	Promotes mucin production [149]

Carboxymethyl cellulose (CMC), ethoxylated sorbitan esters 80 (P80), carrageenan (CGN), Allura red (AR), titanium dioxide (TiO_2), monosodium glutamate (MSG), benzoate (Benz), propionate (Prop), sorbate (Sorb), sulphite (Sulp), sodium stearoyl lactylate (SSL), sophorolipids (SLs), rhamnolipids (RLs), short-chain fatty acid (SCFA), trimethylamine (TMA), colorectal cancer (CRC), non-alcoholic fatty liver disease (NAFLD), metabolic syndrome (MetS).

research in the last two decades has demonstrated how frequently consumed emulsifiers may be promoting IBD and MetS via multiple mode of actions. Emulsifiers such as P80 and CMC can cause dysbiosis, promoting the growth of pathogenic bacteria, whilst also increasing the virulence of pathogenic bacteria [20,44,59]. These two emulsifiers may also disrupt the mucus layer [46,61], whilst multiple emulsifiers have shown to decrease tight junctions, thus weakening the intestinal barrier [42,47,61,65]. Furthermore, it has been demonstrated that the emulsifier CGN activates the same inflammatory pathway as the bacterial toxin LPS [40,50]. It is therefore understandable why it has been hypothesized that emulsifiers might be the primary dietary factor responsible for the rise in CD [163], and why the EFSA identified emulsifiers as the only food additive of concern as part of its emerging risk report in 2015 [164].

Other identified food additives of concern are non-natural colours, flavourings, and preservatives. The primary colour of concern is the whitening agent TiO_2, due to its ability to promote dysbiosis, inflammation and aberrant crypt foci in the colon [86–91]**.** The only flavouring of concern reviewed was MSG. Multiple studies have shown that MSG can promote MetS in both murine models [21,105–108] and human trials [98]. Finally, multiple additives of concern were identified from the preservative category. The identified preservatives promoted gut dysbiosis and impaired glucose metabolism [24,125,128]. It is important to note at this point that a consistent criticism of these studies is that the research has been conducted not administering the additives using realistic human exposure conditions. These include using human equivalent doses, choosing food grade additives, and oral administration of additives via different media. It is imperative in order to establish a balanced

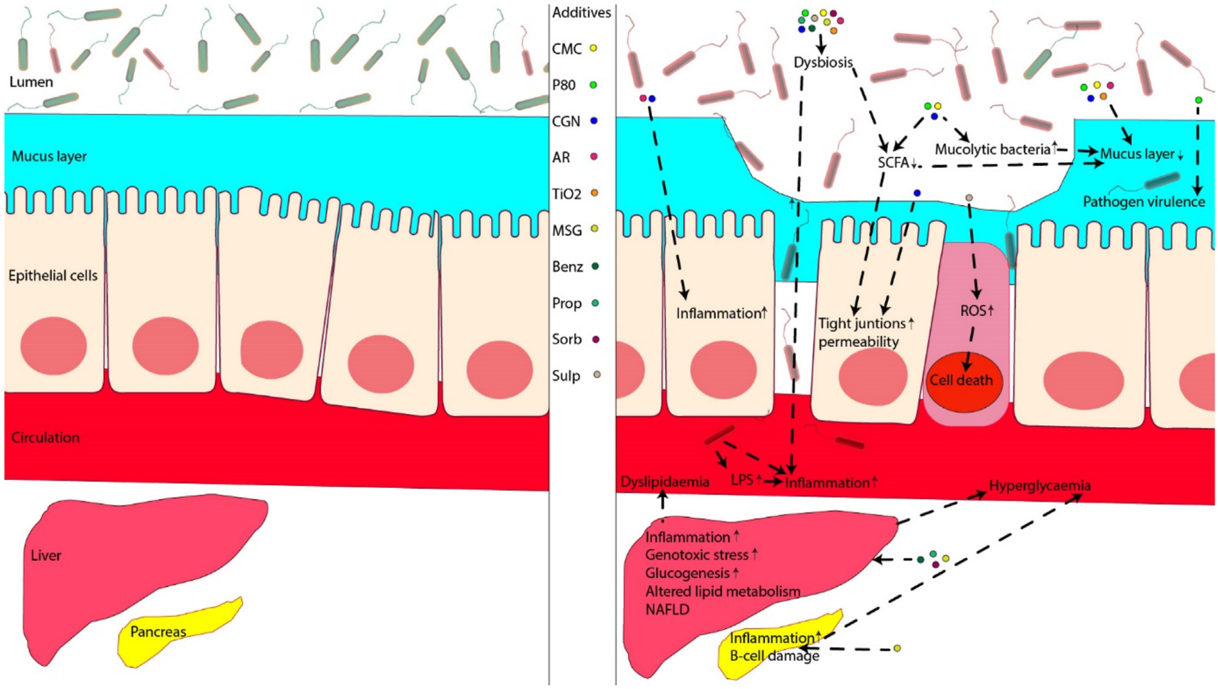

Fig. 2. Image on the left shows a healthy gut environment. Image on the right shows the direct impact of ten common food additives on various parameters associated with inflammatory bowel disease and metabolic syndrome. Carboxymethyl cellulose (CMC), ethoxylated sorbitan esters 80 (P80), carrageenan (CGN), Allura red (AR), titanium dioxide (TiO_2), monosodium glutamate (MSG), benzoate (Benz), propionate (Prop), sorbate (Sorb), sulphite (Sulp), short-chain fatty acid (SCFA), reactive oxygen species (ROS), lipopolysaccharide (LPS), non-alcoholic fatty liver disease (NAFLD). (Made in © Adobe illustrator 2024). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

evidence-based regime that future research should take these criticisms into consideration when designing exposure protocols.

Of note, whilst no studies on the potential synergistic effects of food additives have been reviewed, it is an important field of enquiry. The shortage of such studies is a major limitation in determining whether food additives are complicit in IBD and Mets, and has been mentioned in several recent reviews [33,57,70,136]. Testing the effect of individual food additives might not realistically demonstrate their role in the WD, since the average high-income country consumer ingests multiple food additives every day [19,26,35] and an additive's effect may change when it interacts with other additives [118,125,165,166]. For example, the preservatives benzoate, sorbate and nitrite can inhibit the growth of the probiotic *Clostridium tyrobutyricum* when used together but not individually [123]. Moreover, several different mechanisms of action have been highlighted in this review. One food additive may promote bacterial virulence, another increase the growth of pathogenic bacteria, a third decrease beneficial bacteria and a fourth increase gut permeability [24,42,59] (Fig. 2). Their cumulative effect in promoting disease may be greater than that predicted by their individual effect. Future studies need to investigate the chronic exposure of multiple food additives, using realistic human exposure conditions, to determine whether food additive consumption is contributing to the rise of IBD and Mets.

Credit authorship contribution statement

Darislav Besedin: Conceptualization, investigation, writing – original draft, review, and editing. Rajaraman Eri: Conceptualization, review, and editing. Thi Thu Hao Van: Review and editing. Charles Brennan: Review and editing. Elena Panzeri: Review and editing. Rohan Shah: Review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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